

Senior Data Engineer Take-Home assignment (2-Day) – Project Brief

Setup Overview

Each candidate will be provided with access to an **individual AWS server** to complete this project.

- **Dataset:** Download one of the categories of the Amazon Product Reviews dataset (e.g. the [All_Beauty](#) category) from the official [UCSD repository](#).
- **Database:** Use this [docker-compose.yml template](#) to set up ClickHouse. For how to install docker on Ubuntu, see [here](#). You must:
 - Expose native and HTTP ports
 - Mount volumes
 - Create a user `analytic` with a password stored in a `.env` file (**do not** hardcode credentials)
- **Python Environment:** Set up a Python 3.10 virtual environment for the project.
- **Utilities:** Clone and install the [dbutils](#) package (see [README](#) in the repo).

Tasks

1. Data Ingestion:

- Define a schema called `amazon` in ClickHouse to store the Amazon reviews.
- Create a ClickHouse table called `reviews` with appropriate columns and data types matching the dataset fields – for instance: `user_id` (STRING), `helpful_votes` (INT), etc. Use a suitable engine for the table to enable fast analytics. Consider ordering/primary key that could benefit your queries. For details about the columns, check the data fields section [here](#).
- Write a Python script that **reads the review data in batches** and inserts it into the `reviews` table. The script should ensure that no duplicates are inserted. The script should do a concurrency test before inserting on what observations have already been inserted. Your keys should help a lot here. **Make sure** to use the `dbutils` library for database connectivity and writing/ingesting to the database.
- Add logging to your script:
 - Log when ingestion starts and finishes
 - Log each batch processed (e.g. number of rows inserted)
 - Log any errors encountered during ingestion

2. Analysis & Visualization (SQL + Polars):

- Using the `dbutils` package to fetch data, perform **EDA** to extract relevant aggregates, distributions, and patterns.
- Leverage Polars to further **analyze and visualize** the results.
- Identify **meaningful insights** in the `reviews` data and present them clearly with code and visualizations.

3. Automation Challenge:

- Describe how you would **automate ingestion** by detecting new files in a local `data/` folder and importing them into ClickHouse without duplication.
- This is a conceptual task – no need to implement it, but clearly explain the approach (tools/technologies or pseudocode) you would use to achieve automated data ingestion for continuously arriving files.

Deliverables

- **Code:** Python scripts/notebooks for data ingestion and analysis (well-documented and formatted).
- **SQL Queries:** A list of all SQL queries used for data exploration (can be included in the Python script or a separate `.sql` file).
- A small presentation of the insights from the analysis you did in task 2. This could be in any format you chose. e.g. PDF, Power Point, etc.
- **Automation Write-Up:** A brief write-up detailing your ingestion automation approach (from Task 3). This can be in **Markdown, plain text, PDF, or any other readable format** — whichever you're most comfortable with.
- **README:** A clear README file with instructions to set up the environment (Docker services, virtual environment, etc.), run the ingestion script, execute analyses, and view the results. Include any assumptions and how to reproduce your work.

Submission Instructions

Once completed, please **send a zipped folder** of your project to `jninsiima@bnr.rw` and `cc jnyirimana@bnr.rw` and `dntirushwamaboko@bnr.rw` by **Monday at 09:00AM**.

Deadline: **Failure to adhere to the above deadline will lead to a disqualification.** The servers will be shut down on Monday at 09:00 AM. All the work on the server will be erased.

Helpful Hints

- Functions like `sql_write` expects a **Polars DataFrame**, not a pandas one.
- If you get an error like: `AttributeError: 'dict' object has no attribute 'encode'` ...it may be due to attempting to write a **non-string column** (e.g., an image or metadata object). You might need to **cast that column to str** before writing to ClickHouse.

Questions or Issues?

Join the support WhatsApp group here:

<https://chat.whatsapp.com/IxShxePYEGLCqDLoMQtu4d>

Use the group for any clarifications or issues during the 2-day assignment window.

References

- [Visualization - Polars user guide](#)
- [Installing docker on Ubuntu](#)